

Unit 2 : Project Analysis

1. Present Worth / Net Present Value (NPV)

$$PV = \frac{FV}{(1+r)^n} \quad \blacksquare \quad NPV = \frac{FV}{(1+r)^n} - \text{Initial Cost}$$

where r = discount rate, n = number of years, FV = future cash flow

2. Future Worth (FW)

$$FV = PV \times (1+r)^n$$

r = interest rate, n = years

3. Annual Worth (AW)

$$AW = \frac{NPV}{\text{Annuity Factor}}$$

Annuity Factor = $[1 - (1+r)^{-n}] / r$

4. Internal Rate of Return (IRR)

NPV = 0 ■ solve for r

$$IRR = r_1 + \frac{NPV_1}{NPV_1 - NPV_2} \times (r_2 - r_1)$$

r_1, r_2 = two trial rates where NPV changes sign | Accept if $IRR > \text{cost of capital}$

5. Benefit-Cost Ratio (BCR)

$$BCR = \frac{\text{PV of Benefits}}{\text{PV of Costs}}$$

$BCR > 1 \rightarrow \text{project is beneficial}$ | $BCR < 1 \rightarrow \text{reject project}$

6. Uniform Gradient Cash Flow (G)

$$PV_G = G \times \frac{1}{r} \times \frac{(1+r)^n - 1 - n \cdot r}{r \cdot (1+r)^n}$$

G = constant gradient increase per period | Used when cash flows increase by G each year

7. Return on Investment (ROI)

$$ROI = \frac{\text{Profit}}{\text{Total Investment}} \times 100\%$$

Profit = Total Benefits – Total Costs

Unit 3 : Activity Planning and Scheduling

CPM — Critical Path Method

ES = max(EF of all predecessors)

EF = ES + Duration

LF = min(LS of all successors)

LS = LF – Duration

Slack (Float) = LS – ES

also: Slack = LF – EF

Critical Path $\rightarrow$ all activities where Slack = 0

PERT — Expected Time (t_e)

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

t_o = optimistic | t_m = most likely | t_p = pessimistic

Variance (σ^2)

$$\sigma^2 = \left[\frac{t_p - t_o}{6} \right]^2$$

Standard Deviation (σ)

$$\sigma = \sqrt{\sigma^2} = \frac{t_p - t_o}{6}$$

PDM — Precedence Diagramming Method (Lag / Lead)

$$\text{Finish-to-Start: } ES_{\text{successor}} = EF_{\text{predecessor}} + \text{Lag}$$

$$\text{Float} = LS - ES = LF - EF$$

Lag > 0 → delay | Lead (negative lag) → overlap | Used to shorten project duration

Unit 4 : Risk Management**Z-value Formula (Schedule Risk)**

$$Z = \frac{T_s - T_e}{\sigma}$$

T_s = scheduled/target time | T_e = expected (PERT) time | σ = standard deviation of project

Look up Z in Normal Distribution table → gives probability of finishing on time

Unit 6 : Monitoring and Control**Earned Value Analysis (EVA)**

$$BCWP (EV) = \% \text{ Complete} \times BAC$$

$$BCWS (PV) = \text{Planned \%} \times BAC$$

$$CV = EV - AC$$

$$SV = EV - PV$$

$$CPI = \frac{EV}{AC} \quad SPI = \frac{EV}{PV}$$

EV = Earned Value | AC = Actual Cost | PV = Planned Value | BAC = Budget at Completion

CPI > 1 = under budget | SPI > 1 = ahead of schedule

Estimate at Completion (EAC)

$$EAC = \frac{BAC}{CPI} \text{ or } EAC = AC + (BAC - EV)$$

PERT vs CPM

PERT	CPM
Event-oriented	Activity-oriented
Probabilistic	Deterministic
Time uncertain	Time known
Focus on time	Focus on time & cost

NPV vs IRR

NPV	IRR
Actual profit value	Percentage return
Uses discount rate	Finds discount rate
Better for large projects	Useful for investment comparison

Verification vs Validation

Verification	Validation
Building the product right	Building the right product
Process-oriented	Product-oriented
During development	After development

QA vs QC

QA	QC
Prevents defects	Detects defects
Process-focused	Product-focused
Proactive	Reactive

Important Testing Levels

1. Unit Testing
2. Integration Testing
3. System Testing
4. Acceptance Testing

Important Testing Types

- Black Box Testing
- Beta Testing
- Regression Testing
- Smoke Testing
- Alpha Testing
- White Box Testing